

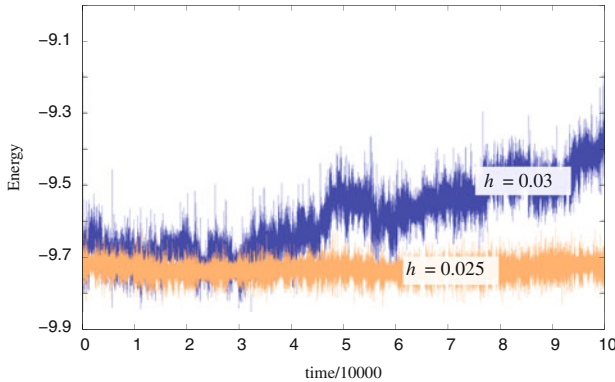


Fig. 3.10 Energy as a function of time for the Verlet method with two different stepsizes. The drift is controlled for $h = 0.025$ but much more dramatic at $h = 0.03$

is not well conserved by the Verlet method. The explanation is that the backward error analysis presented previously, and therefore the assumption that the numerical method is described by a Hamiltonian flow map, is invalid. This is a consequence of the mentioned fact that construction of the modified Hamiltonian depends on an optimal truncation at high order. For $h = 0.03$ in this example, the truncation would introduce a substantial error, or, alternatively, the “Hamiltonian flow” interpretation for the numerical method is poor. Either the stepsize must be reduced in such cases (already $h = 0.025$ shows much less drift) or some other method must be used.

3.5 Energy Conserving Methods

It is possible to derive numerical methods that exactly (i.e. to within rounding error) conserve the energy at each step and this may seem a more natural approach to molecular dynamics than symplectic integration since we have discussed and evaluated methods in terms of their energy conservation.

Consider the following 2D example of a differential equation system

$$\dot{u} = f(u, v), \quad \dot{v} = f(u, v).$$

Notice that this system has a first integral

$$I(u, v) = u - v,$$

and consider the application of Euler’s method:

$$u_{n+1} = u_n + hf(u_n, v_n), \quad v_{n+1} = v_n + hf(u_n, v_n).$$